

Timing Risk

$$TR(r_k) = \sqrt{\sum_{k=1}^n r'_k C r_k} \quad (9.4)$$

where C is the trading risk covariance matrix expressed in $(\$/\text{share})^2$, and is scaled for the length of the trading interval, and r_k is the residual vector of unexecuted shares at the beginning of period k .

One-Sided Optimization Problem

It is important to note that our I-Star market impact equation (Equation 9.1) requires that the trade size X be positive, i.e., $X_i > 0$ for all stocks. This creates a difficulty when optimizing a two-sided portfolio since we need to have a way to incorporate the negative market relationship between buy and sell orders. Similar to how we adjusted the price appreciation term by the side of the order, we adjust the covariance term by the side of each order and thereby convert the portfolio optimization problem to a one-sided optimization problem. That is:

$$c_{ij}^* = \text{side}(i) \cdot \text{side}(j) \cdot c_{ij}$$

Then our transaction cost equations will properly account for the sided covariance across stocks. Notice in this calculation that the variance of a stock will be positive, the covariance between two stocks on the same side (i.e., both buys or both sells) will be equal to the original covariance term and the covariance between two stocks with opposite side orders (i.e., one buy and one sell) will be the negative of the original covariance term.

OPTIMIZATION FORMULATION

Using the expressions above the complete portfolio trader's dilemma translates to:

$$\text{Min}_x \left\{ \sum_{i=1}^m \left(\sum_{t=1}^n \frac{b_1 \cdot I_i^* \cdot x_{it}^2}{X_i \cdot v_{it}} \right) + (1 - b_1) I_i^* + \sum_{i=1}^n \sum_{t=1}^m x_{ij} \cdot \Delta p_{ij} \cdot t \right\} + \lambda \cdot \sqrt{\sum_{k=1}^n r'_k C r_k} \quad (9.5)$$

Subject to constraints:

	Completion
i. $\sum_{t=1}^n x_{it} = X_i$	
ii. $x_{it} \geq 0$	No Short Sales
iii. $r_{it} - r_{it+1} \geq 0$	Shrinking Portfolio
iv. $r_{it} = \sum_{k=t}^n x_{ik}$	Residual Schedule